

When we use the formula:

$$y_2 = y_1 \int \frac{e^{-\int P(x)dx}}{y_1^2} dx$$

We don't need integration constants.

$$y_2 = y_1 \int \frac{e^{-\int P(x)dx + C}}{y_1^2} dx$$

$$y_2 = e^C y_1 \int \frac{e^{-\int P(x)dx}}{y_1^2} dx$$

The e^C is a number that scales y_2 .

choosing $C=0$ makes $e^C = e^0 = 1$.

$$y = C_1 y_1 + C_2 y_2$$

the e^C can be absorbed in the C_1 .

$$y_2 = y_1 \left[\int \frac{e^{-\int P(x) dx}}{y_1^2} dx + C \right]$$

$$y_2 = y_1 \int \frac{e^{-\int P(x) dx}}{y_1^2} dx + \textcircled{C y_1}$$

the $C y_1$ is not needed.

We already have: $\boxed{C_1 y_1}$

in the general solution:

$$y = \boxed{C_1 y_1} + C_2 y_2$$
